Inborn Errors of Immunity

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Michael H. Albert, Arjan Lankester,
Andrew Gennery, and Bénédicte Neven

90.1 Introduction

Inborn errors of immunity (IEI) arise from genetic defects that lead to abnormalities in immune cell development or function with a wide spectrum in severity and clinical manifestations. These include infectious, autoimmune, autoinflammatory, hematological, and malignant presentations. Allo-HCT provides a life-saving and curative treatment modality in many but not all of these disorders by replacing the defective hematopoietic cell lineage(s) by those from healthy allogeneic donors. Other management options including enzyme replacement therapy, gene transfer into autologous hematopoietic stem cells, and targeted therapies (see below) may provide an alternative or bridging approach to HCT in specific IEI.

90.2 Diseases

IEI may be broadly categorized into severe combined immunodeficiencies (SCID) and non-SCID. To further subdivide non-SCID, the phenotypic classification as suggested by the International Union of Immunological Societies (IUIS) Inborn Errors of Immunity Committee can be used, which encompasses >500 genetic causes of IEI (Table 90.1).

Table 90.1 Phenotypic classification of IEI as suggested by the International Union of Immunological Societies (IUIS) Inborn Errors of Immunity Committee (Bousfiha et al. 2022)

1.	Immunodeficiencies affecting cellular and humoral immunity (CID)
2.	CID with associated or syndromic features
3.	Predominantly antibody deficiencies
4.	Diseases of immune dysregulation
5.	Congenital defects of phagocyte number, function, or both
6.	Defects in intrinsic and innate immunity
7.	Auto-inflammatory disorders
8.	Complement deficiencies
9.	Bone marrow failures
10.	Phenocopies of IEI

M. H. Albert (✉)
Pediatric SCT Program, Dr. von Hauner University Children's Hospital, Ludwig-Maximilians Universität, Munich, Germany
e-mail: michael.albert@med.uni-muenchen.de

A. Lankester
Department of Pediatrics, Stem Cell Transplantation Program, Willem-Alexander Children's Hospital, Leiden University Medical Center, Leiden, The Netherlands
e-mail: a.lankester@lumc.nl

A. Gennery
Paediatric Immunology + HSCT, Great North Children's Hospital, Newcastle University, Newcastle upon Tyne, UK
e-mail: andrew.gennery@newcastle.ac.uk

B. Neven
Pediatric Immune-Hematology and Rheumatology Unit, Necker Children Hospital, Assistance Publique Hopitaux de Paris, Paris City University, Paris, France
e-mail: benedicte.neven@aphp.fr

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Updated guidelines for HCT of IEI together with detailed protocols have been published by the EBMT Inborn Errors Working Party (EBMT IEWP) in 2021 (Lankester et al. 2021) and can be accessed at https://www.ebmt.org/sites/default/files/2021-07/EBMT%20ESID%20IEWP%20Guidelines%20for%20HCT%20for%20inborn%20errors%20of%20immunity.pdf

These guidelines include detailed recommendations for the use of six conditioning regimens with varying degrees of intensity depending on type of IEI, patient condition, and age.

90.3 SCID

The overall frequency of SCID was for a long time estimated to be 1 in 50,000–100,000 live births. However, in recent years, newborn screening programs making use of the T-cell receptor excision circles (TREC) technology have demonstrated that the frequency may actually be two- or more-fold higher with clear geographical and ethnic differences (Kwan et al. 2014; Rechavi et al. 2017; Speckmann et al. 2023).

The immunological phenotypes of SCID are shown in Table 90.2 representing monogenic inherited defects in T-, B-, and NK-cell differentiation leading to the absence or inactivity of corresponding mature cells.

In the absence of newborn screening programs or positive family history, most patients present within the first 3–6 months with severe, recurrent, or opportunistic, often life-threatening, infections, the most common being *Pneumocystis jiroveci pneumonia*. Other common symptoms include diarrhea, dermatitis, and failure to thrive. Survival in SCID patients depends on expeditious T-cell reconstitution, and in the absence of successful HCT, or autologous stem cell gene therapy, most children usually die during the first year of life. As many as 50% of SCID patients are engrafted with maternal T-cells but in most instances these cells do not initiate GvHD. Transfusion-associated GvHD, on the other hand, is frequently lethal in SCID, and any patient with a possible diagnosis of SCID must receive irradiated blood products. Bacille Calmette–Guérin (BCG) vaccination can give rise to disseminated BCG-osis, and rotavirus vaccine can cause persistent enteritis in SCID patients, and all live vaccines should be avoided if there is any suspicion, a positive NBS result, or a positive family history of severe T cell deficiency. With the ongoing introduction of NBS in Europe, the presentation of SCID patient is about to change to mostly healthy-appearing newborns without signs of active infection. However, not all peri- and early postnatal infections like, i.e., CMV will be avoidable in infants identified by NBS (Speckmann et al. 2023; Thakar et al. 2023).

Table 90.2 Gene defects typically associated with specific SCID phenotypes

T-B + NK−	T-B + NK+	T-B − NK−	T-B − NK+
IL2RG (SCID-X1)	*IL7R*	*ADA*	*LIG4*
JAK3	*CD3D*	*AK2* (reticular dysgenesis)	*RAG1*
	CD3E	*RAC2*	*RAG2*
	CD247 (CD3ζ)		*DCLRE1C* (Artemis def.)
	CORO1A *ITPKB* *LAT*		*NHEJ1* (Cernunnos XLF)
	PTPRC (CD45 def.) *SLP76* Thymic defects* *FOXN1* *PAX1* Complete DiGeorge syndrome CHARGE syndrome		*PRKDC* (DNA-PKcs def.)

Adapted from Bousfiha et al. 2022. *: these may present with a SCID phenotype or at newborn screening, but HCT may be contraindicated or not the primary treatment choice

90.3.1 General Principles in Allo-HCT for SCID

The Stem CEll Transplant for primary Immune Deficiencies in Europe (SCETIDE) registry has collected data on SCID transplants comprising 50 years of HCT experience, and a number of

important publications have documented the outcomes and important risk factors (Fischer et al. 1990; Antoine et al. 2003; Gennery et al. 2010; Lankester et al. 2022). Recently, studies from the North American group have reported similar findings (Pai et al. 2014; Heimall et al. 2017, Thakar et al. 2023). The major factors positively influencing outcome reported in these studies include:

1. Absence of infections at HCT
2. Age at transplant <3.5 months or diagnosis via NBS
3. Use of MFD and MUD
4. Use of myeloablative conditioning
5. Recovery of naive CD4 T lymphocytes >0.5 × 10^3/μL at +1 year after HCT

90.3.2 Donor, Graft Source, and Conditioning in HCT for SCID

Overall survival for MD (related or unrelated) HCT in SCID has improved to over 90% (Lankester et al. 2022; Thakar et al. 2023). Virtually all infants have a haploidentical parental donor, and this is an alternative option, as the donor is readily available. HLA disparity necessitates rigorous in vitro or in vivo TCD in order to reduce to risk of GvHD. Most centers now employ either TCR alpha/beta depletion (Balashov et al. 2015; Shah et al. 2018) or PT-CY (Neven et al. 2019). Although promising survival rates have been reported, longer follow-up in a larger cohort of patients is required to determine the position of these approaches. CD34+ selection of the graft also remains an option in selected cases. CB can be a suitable alternative stem cell source (Fernandes et al. 2012). Theoretical advantages for using CB stem cells include rapid availability, less risk of GvHD compared to adult URD, no medical risk to the donor, and a greater proliferative life span which might be particularly important in such young recipients. There are some specific disadvantages including sometimes slower engraftment, lack of viral-specific cytotoxic T-cells, and lack of availability of the donor for a boost HCT.

Although HLA-genoidentical sibling donor BM may be infused into SCID recipients without any conditioning or GvHD prophylaxis, usually only T-cells of donor origin will engraft, and myeloid and often B-cells will remain of recipient origin. Patients with insufficient myeloid engraftment and/or poor or declining naïve T-cell compartments may experience severe complications that require a second, conditioned HCT (Riller et al. 2023). Therefore, and if tolerable for the patient, conditioning is recommended for all SCID patients in order to achieve optimal clinical and immunological outcomes (Lankester et al. 2022).

Individualized approaches making use of therapeutic drug monitoring provide novel and less toxic options to improve HCT outcome in these vulnerable young infants, while antibody-based conditioning approaches that may limit systemic toxicity are currently being developed.

90.3.3 Omenn's Syndrome

Omenn's syndrome (OS) is characterized by SCID typically associated with the triad of erythroderma, hepatosplenomegaly, and lymphadenopathy. There is a marked eosinophilia and a variable number of autologous, activated, and oligoclonal T lymphocytes, which infiltrate target organs and are generally poorly responsive to mitogens. Whereas outcomes in HCT for OS were traditionally inferior compared to classical SCID, results have improved in recent years (Gennery et al. 2010; Heimall et al. 2017; Thakar et al. 2023). The overall mortality in these studies was lower than previously reported and was due to early recognition of OS and rapid initiation of treatment with topical/systemic immune suppression with steroids, cyclosporin A, and T-cell directed serotherapy to control immune dysreactivity before proceeding to HCT.

90.4 Non-SCID IEI

The landscape of HCT in non-SCID IEI has dramatically changed over the last decade

- New genetic causes of IEI are being described in accelerating frequency thanks to next-

generation sequencing techniques (Bousfiha et al. 2022). Functional consequences of genetic variants in the same gene can be broad, from complete or partial loss to gain of function resulting in different clinical phenotypes. Somatic variants may also occur, mimicking disease caused by germline variants.

- The concept of "pure" immunodeficiencies with predisposition to infections has been abandoned with many newly described autoimmune, auto-inflammatory conditions, or syndromal disorders with immunodeficiency. Many of these diseases can be cured by HCT, while in syndromal disorders only the hematopoietic portion of the disease can be corrected, which may nevertheless be indicated and result in not just increased survival but also quality of life in selected patients.
- HCT outcomes have further improved with around 90% OS and low GVHD rates after MSD or MUD HCT in many non-SCID IEI, like, i.e., CGD and WAS (Chiesa et al. 2020; Albert et al. 2022b).
- In general, donor availability has become less of a factor in the decision-making process to go for HCT. Current outcomes of MFD and MUD are highly similar, while results with mismatched/haplo-identical HCT are gradually approaching those obtained with MD especially when performed in experienced centers (Shah et al. 2018; Neven et al. 2019; Kurzay et al. 2019).
- The importance of DFS and GRFS as compared to OS is increasingly appreciated and addressed in medium- to long-term outcome studies, also in comparison to non-HCT approaches (Speckmann et al. 2017; Barzaghi et al. 2018; Albert et al. 2022b).
- More IEI patients are discovered with very mild or atypical phenotypes of well-known IEI, and these often hypomorphic genetic variants are especially challenging with respect to timely recognition and management (Notarangelo et al. 2016; Schuetz et al. 2023). Especially in these "milder" cases, often recognized in adolescence and adulthood, quality of life is increasingly a factor in HCT decision making (Cole et al. 2013; Cheminant et al. 2023).
- Adolescents and young adults with IEI are increasingly appreciated as candidates for HCT, and outcomes are encouragingly good (Albert et al. 2018; Fox et al. 2018), with survival determined by the underlying IEI entity, pre HCT comorbidities, and organ damage rather than age at HCT or donor type (Albert et al. 2022a).

The consequence of these developments has been that many more patients with IEI are today considered for, referred for, and counselled about HCT.

90.4.1 Indication

One of the most challenging aspects of transplant for non-SCID IEI is the question of whether a high-intensity, curative therapy with HCT is indicated. For some IEI (like FHLH), the immediate HCT indication is clear, while for many other IEI this will be less evident because of the high interindividual variability of the clinical phenotype. In diseases where long-term prognosis is known to be poor, even presymptomatic patients can be subjected to HCT (like i.e. X-CGD). A genetic diagnosis is increasingly made in IEI patients and may support the decision to proceed to HCT. Similarly, a positive family history with severe disease manifestations or a known poor long-term prognosis of the disease may justify early preemptive HCT. Still, given the highly variable and frequently unpredictable disease course in IEI, a genetic diagnosis alone without clinical manifestations is in many IEI an insufficient reason to perform HCT. In all IEI patients, the decision to proceed with HCT should always include interdisciplinary discussions with patients and their families, also addressing issues like fertility/family-planning, school/work, other psycosocial factors, and quality of life.

90.4.2 Conditioning (Table 90.3)

In IEI, it is the goal is to establish sufficient long-term donor chimerism in the affected cell lineage, while reducing short- and long-term toxicity to a minimum. The required degree of donor chimerism for effective and sustainable disease correction varies depending on the type of IEI and has not yet been established for all entities. Accordingly, the graft source, serotherapy, and type of conditioning regimen should be chosen depending on many factors, including IEI entity, age of patient, donor type, active infections, comorbidities, organ damage, and experience of the center.

Conventional MAC preparation with BU-/CY-based regimens has historically been associated with significant treatment-related toxicity and TRM. The IEWP of EBMT had begun in 2005 to publish detailed recommendations for conditioning of IEI as discussed above (Lankester et al. 2021). These recommendations include:

1. Replacement of CY with FLU, as the combination of BU/FLU with pharmacokinetic targeting of BU AUC appears to be better tolerated in these patients.
2. The option to replace BU with a structural analogue, TREO, which is similarly immuno- and myelosuppressive but causes less hepatic SOS/VOD (Slatter et al. 2018).
3. Addition of a submyeloablative dosing regimen for BU/FLU (Güngör et al. 2014).
4. Establishing RIC to achieve stable engraftment of immunocompetent donor cells with reduced procedure-related morbidity and mortality (Veys 2010).
5. An alkylator-low regimen for radiosensitive disorders (Slack et al. 2018).

Table 90.3 Conditioning regimens recommended by IEWP (Lankester et al. 2021)

Regimen	
A	BU AUC 85–95 mg × h/L, FLU 160 mg/m^2
B	TREO 30–42 g/m^2, FLU 150–160 mg/m^2, TT 8–10 mg/kg
C	BU AUC 60–70 mg × h/L, FLU 160–180 mg/m^2
D	TREO 30–42 g/m^2, FLU 150–160 mg/m^2
E	FLU 150–160 mg/m^2, MEL 140 mg/m^2
F	FLU 150 mg/m^2, CY 20–40 mg/kg

90.5 HCT for Radiosensitive IEI

Patients with radiosensitive IEI such as Nijmegen-breakage syndrome, DNA ligase 4 deficiency, Cernunnos deficiency, or DNA-PKcs deficiency including those with a SCID phenotype are increasingly being identified and considered for HCT. As many of the conditioning regimens are particularly damaging to DNA, less toxic regimens are required to successfully treat these patients (Slack et al. 2018). No definitive studies are available, but low-dose FLU/CY regimens like the one suggested by the EBMT IEWP guidelines have been proven effective and safe (Lankester et al. 2021).

90.6 Alternative Therapies

Alternative treatments to HCT have been developed for specific IEI over the last three decades.

90.6.1 Enzyme Replacement Therapy (ERT) for Adenosine Deaminase Deficiency (ADA-SCID)

Enzyme replacement in ADA deficiency with PEG-ADA is administered weekly or twice weekly by IM injection and leads to rapid metabolic correction which is followed by cellular and humoral immune reconstitution. The extent of immune recovery is variable, and a significant number (~50%) remain on Ig replacement. Over a longer time period, patients show a decline in T-cell numbers and remain lymphopenic. Long-term follow-up shows that patients may remain clinically well, but clinical problems can arise and a number of cases of EBV-related lymphoma have been reported (Chan et al. 2005). Given the

improved outcomes of HCT and GT in recent times, ERT is predominantly considered as a bridge to stem cell-based curative therapy.

90.6.2 Gene Therapy for Specific IEI

Autologous stem cell gene therapy (GT) via vector-mediated transfer of healthy copies of an affected gene into autologous $CD34^+$ cells has progressed from a highly experimental therapy to the first licensed gene therapy for an IEI (ADA-SCID) within the last two decades. One of the major conceptual advantages of GT is the elimination of the inherent risk of GVHD associated with allogeneic HCT procedures.

Clinical trials performed with gamma retroviral vectors for ADA-SCID, X-linked SCID (SCID-X1), chronic granulomatous disease (CGD), and Wiskott-Aldrich syndrome (WAS) demonstrated that gene therapy can be an effective treatment option in patients lacking an HLA-identical donor (Hacein-Bey-Abina et al. 2002; Boztug et al. 2010; Stein et al. 2010; Aiuti et al. 2009). However, a high rate of insertional mutagenesis was observed in trials for SCID-X1, WAS, and CGD (Ott et al. 2006; Hacein-Bey-Abina et al. 2003; Braun et al. 2014). This has prompted the development of safer vectors based on self-inactivating retroviral or lentiviral vectors. Currently, a number of trials are ongoing or concluded for a number of IEI. All share the concept of submyeloablative conditioning followed by the infusion of autologous stem/progenitor cells transduced with the wildtype gene. Promising results were published, especially for ADA-SCID (Kohn et al. 2021; Cicalese et al. 2016), WAS (Ferrua et al. 2019) and SCID-X1 (Mamcarz et al. 2019). It is expected that gene editing approaches as an alternative for gene addition technologies will be developed for stem cell based GT in the next few years and may potentially also be employed to correct mature cells in diseases like, i.e., CD40 ligand deficiency, CTLA4, and IPEX.

In theory, autologous stem cell gene therapy offers the appealing prospect of avoiding allo-immune reactions such as GVHD or rejection and a lower conditioning-related toxicity compared to allo-HCT. But its exact role in treatment algorithms still needs to be defined in the absence of comparative studies. Also, logistic, regulatory, and economic hurdles still have to be overcome before its widespread application in the treatment of IEI. Nevertheless, it has widened the therapeutic repertoire for patients with some IEI. The rapid evolution of novel gene correction approaches has the potential to lead to even safer and more effective treatment options.

90.6.3 Targeted Therapies

The unravelling of new genetic IEI entities, especially those caused by gain-of-function (GOF) variants and their pathophysiology, has for the first time opened the possibility to treat these diseases with highly specific, often small molecule inhibitors, some of which are already approved for other indications. These include but are not limited to abatacept for CTLA4 haploinsufficiency, ruxolitinib and other JAK-inhibitors for STAT1 GOF, leniolisib for APDS, etanercept for ADA2 deficiency, and IL-1-targeted therapies (anakinra, rilonacept, and canakinumab) for auto-inflammatory recurrent fever syndromes (Ochs and Petroni 2018; Ombrello et al. 2019; Rao et al. 2023). We can expect that more substances (like, i.e., IL-18BP for XIAP) may be licensed or repurposed for the treatment of IEI in the future. At this point in time, the exact role of these agents in the treatment algorithm of IEI is unclear. Ideally, they could make HCT unnecessary for some patients. On the other hand, concerns about long-term toxicity, infection risk, and lymphoma risk exist. In any case, in some patients with excessive autoimmunity and/or inflammation, these therapies can be viewed as an ideal bridge to HCT and considered as a remission induction strategy to control the underlying IEI, because they have the potential to bring the patient into the best possible clinical condition for HCT.

Key Points

- IEI require a tailored approach to HCT management, and disease-specific transplant protocols have been developed for these diseases, including stem cell-based GT.
- Preceding comorbidity, particularly infectious complications at HCT and concurrent end-organ damage, adversely affects outcome. For many diseases, HCT at an early age is recommended.
- Alternative therapies—often as a bridge to transplant—are increasingly available to improve patient outcomes.

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